SP1 · NONLINEAR-PUSHOVER

Pushover cantilever moment hinge (CSI 1-026, moment-hinge scope)

Production Pushover pipeline — cantilever moment hinge vs CSI 1-026 (moment-hinge scope)

PASS

error −0.0000061% · tol 1%

§2 TECHNICAL FEATURES TESTED

- Production **buildPushoverTcl** run on the **STRIX engine** pushover route (indirect displacement control with event-limited steps)
- FEMA lumped-hinge assembly: hinge spring (I) – elastic body – hinge spring (J)
- Hinge backbone = REAL `convertFemaToModIMK` output (csi −10%Ke clamp)
- Self-consistency check at every recorded step (not exact-P-target crossings)
- One rigid-diaphragm floor at the tip → the production uniform pattern is a single tip force

§3 PROBLEM DESCRIPTION

CSI SAP2000 Software Verification Example 1-026 tests a cantilever with a moment hinge at the fixed end, pushed at the free tip until the hinge yields, hardens to a capping moment, and drops toward a residual plateau. STRIX has no discrete shear hinge (shear is a separate force-controlled check, not part of the pushover hinge itself — devplan §12.7), so this benchmark covers the **moment-hinge backbone only**, reusing CSI's published My/Mc/Mr/θ_C values directly. The model is a 90° axis relabeling of CSI's horizontal cantilever (STRIX's Pushover lateral pattern is inherently horizontal, and the FEMA lumped-hinge branch requires a geometrically vertical element) — a vertical "column" fixed at the base and pushed horizontally at the free tip, with a moment hinge at the base only (the tip hinge never sees moment for a tip-loaded cantilever, so its own backbone is inert). The tip is one rigid-diaphragm floor, so the production uniform lateral pattern is a single force at the tip; a tiny 1 N gravity load gives that floor its mass. With P-Delta off and a moment-only hinge the axial force enters neither the lateral stiffness nor the hinge strength, so it changes none of the checked quantities. CSI fixes the column EI, while production reduces an RC column's flexural stiffness by the effective-stiffness factor (0.30 at this axial-load level); the model therefore inputs I/0.30, taken from the same production function, and the run confirms that the emitted body stiffness equals CSI's EI. Because the pushover driver's step landings are adaptive and cannot be forced onto an exact target load, the check is self-consistency: at every recorded step, statics gives the base moment exactly (M=P×L, independent of any model), the production hinge backbone gives the theoretical hinge rotation for that moment, and the resulting theoretical tip deflection is compared against what STRIX actually recorded.

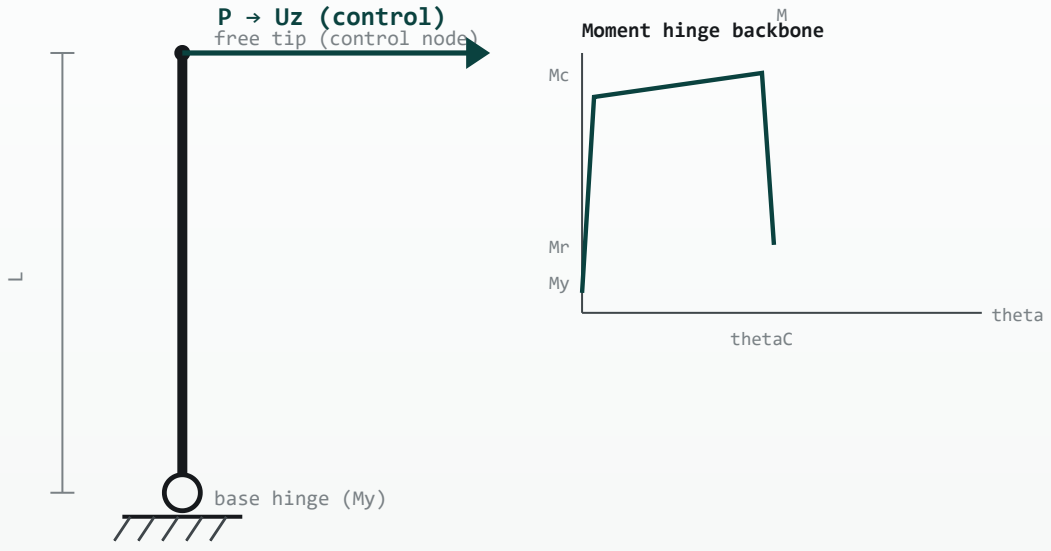


Figure SP1. Vertical cantilever column with a moment hinge at the base, pushed horizontally at the free tip (control node); inset shows the moment-hinge backbone (My→Mc hardening→Mr residual) reused verbatim from CSI 1-026.

§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		MOMENT HINGE BACKBONE		MATERIAL & ANALYSIS	
Length L	24 in	My	1440 kip-in	E	3600 ksi
Section	b12 × d18 in (rect)	Mc	1920 kip-in	thetaY (registered)	0.0025 rad
Inertia I	5832 in⁴	Mr	480 kip-in	Backbone clamp	csi (−10%Ke)
Orientation	vertical (90° relabel)	thetaC	0.04 rad	Engine	STRIX engine (pushover)

§5 REFERENCE SOLUTION

Tip Uz at Point1 (P=My/L, elastic boundary): P*L^3/(3*EImod) + L*theta_hinge(M=My) = 0.071972 in

Tip Uz at Point2 (P=Mc/L, hardening peak): P*L^3/(3*EImod) + L*theta_hinge(M=Mc) = 0.975962 in

Uz self-consistency across ALL recorded pre-peak (ascending) steps: worst |Uz_recorded - Uz_theory(P_recorded)| / Uz_theory over every step up to and including the peak = 0 %

CSI SAP2000 Software Verification Example 1-026 'Frame – Moment and Shear Hinges' (PDF confirmed, devplan §12.7) — moment-hinge backbone (My/Mc/Mr/thetaC) taken verbatim from CSI's published figure. The independent target values below are NOT CSI's own published 'Independent' column (those include a GAv shear term and a 0.4in shear-hinge plastic term with no STRIX counterpart — STRIX has no discrete shear hinge). Re-derived from scratch by sp1Theory()/sp1UzRyAt() using the SAME unit-load method (Cook & Young 1985 p.244, as CSI itself cites) minus the shear terms, with the elastic term using STRIX's actual body stiffness EI_mod=(11/10)×EI (TclBuilder.ts N_FAC=10 series-stiffness correction, RC-column FEMA lumped-hinge branch) and the hinge rotation taken from the REAL production convertFemaToModIMK() clamp output (csi mode, -10%Ke), not assumed.

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Tip Uz at Point1 (P=My/L, elastic boundary) (mm)	step1.ctrlDisp	1.828078	1.828097	−0.0010%	PASS
Base hinge rotation at Point1 (rad)	step1.hingeRotI	0.0025	0.0025	−0.0012%	PASS
Tip Uz at Point2 (P=Mc/L, hardening peak) (mm)	step24.ctrlDisp	24.789437	24.789439	−0.0000061%	PASS
Base hinge rotation at Point2 (= thetaC) (rad)	step24.hingeRotI	0.04	0.04	−0.0000060%	PASS
Uz self-consistency across ALL recorded pre-peak (ascending) steps (%)	worst of 25 steps	−0.001026	0	−0.0010%	PASS

§7 SELF-CONSISTENCY (NO MESH SWEEP — SINGLE ELEMENT)

CHECK	WORST |DIFF| (%)	DIFF
single element (production arc-length driver)	−0.001026	−0.0010%

SP1 uses a single, non-refinable element (the production FEMA lumped-hinge assembly is always one elastic body between two hinge springs, regardless of mesh) — there is no convergence sweep. Instead, every recorded pre-peak step is an independent self-consistency check against the production hinge backbone; the worst deviation across all of them is `about 0.001%`, i.e. near machine precision. The STRIX engine's event-limited steps land exactly on the backbone corners: Point1 at the yield rotation (0.0025 rad) and Point2 at the capping rotation (0.040 rad). Point3 (post-drop residual) is reported as a diagnostic, not gated: the STRIX engine traces the whole descending branch step by step and reaches the residual point exactly (0.065 rad, tip 39.725 mm = theory); beyond it the hinge stays on its flat residual plateau, while the closed form keeps extending the descending slope, so their difference past Point3 is a definition difference, not an error. CSI's own conclusion for this problem notes that such steep hinge drops "should be avoided whenever possible" and that its coarse-mesh model deviates 29–41% from the idealized hand calculation.

§8 CONCLUSION

PASS

The production pushover pipeline on the STRIX engine reproduces the hinge backbone's theoretical response to **about 0.001%** across every recorded pre-peak step (Point1 elastic boundary and Point2 hardening peak both pass at a 1% tolerance), and follows the descending branch to the residual point exactly. The moment-hinge assembly, the −10%Ke slope-limit clamp (since v0.9.291 measured from the capping moment to the residual, which sets the post-cap rotation to 0.025 rad) and the base-reaction recorder are verified against the production code path directly. Point3 remains a diagnostic outside the pass/fail gate.

§9 REFERENCES & ANALYSIS FILES

1. CSI — *SAP2000 Software Verification, Example 1-026:* Frame – Moment and Shear Hinges (moment-hinge backbone reused verbatim; shear hinge out of scope — STRIX has no discrete shear hinge).
2. Cook, R. D. & Young, W. C. — unit-load method (cited by CSI 1-026's own hand calculation), *Advanced Mechanics of Materials*, p.244.
3. STRIX production `buildPushoverTcl` (FEMA lumped-hinge assembly, `convertFemaToModIMK` csi backbone clamp) run on the STRIX engine pushover route; the RC column effective-stiffness factor from `flexuralStiffness.columnFlexuralStiffnessFactor`.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · SP1/` `sha256 (pending publish)`